

Scan Bot Specification – Detailed Version (18-03-2026)

1. Objective

Develop a scan bot that runs once per day after US market close.

The bot generates a shortlist of trade candidates based on predefined technical criteria.

The bot DOES NOT execute trades.

2. Universe Definition

Input: Static list of symbols (user-defined).

No dynamic universe expansion.

Symbols are US stocks and ETFs.

3. Data Requirements

ONLY fully closed candles must be used.

Daily: last completed daily candle (regular session).

Weekly: last fully closed weekly candle.

No intraday or partial data allowed.

Timezone reference: US market (New York).

4. Definitions & Formulas

Close = last daily close price.

Support = lowest price in defined recent range (implementation detail to be agreed).

Resistance = highest price in defined recent range.

$\text{SupportDistance} = (\text{Close} - \text{Support}) / \text{Support}$

$\text{ResistanceDistance} = (\text{Resistance} - \text{Close}) / \text{Close}$

$\text{Amplitude} = (\text{Resistance} - \text{Support}) / \text{Support}$

$\text{DollarVolume} = \text{Close} \times \text{Volume}$

$\text{AvgDollarVolume20} = \text{average}(\text{DollarVolume} \text{ over last 20 days})$

5. Mandatory Filters (ALL must pass)

1. Weekly Trend:

- $\text{Close} > \text{MA200 (weekly)}$
- $\text{MA200 (weekly) trending upward}$

2. Liquidity:

- $\text{AvgDollarVolume20} \geq 20,000,000 \text{ USD}$

3. Amplitude:

- $\text{Amplitude} \geq 10\%$

4. Support Proximity:

- $\text{SupportDistance} \leq 5\%$

5. Resistance Distance:

- $\text{ResistanceDistance} \geq 5\%$

6. Scoring Model (applied AFTER filters)

Total Score = 0 to 10

1. Weekly Trend:

2 = conditions met

0 = otherwise

2. Support Proximity:

2 = $\leq 2\%$

1 = $>2\%$ and $\leq 5\%$

0 = $>5\%$

3. Amplitude:

2 = $\geq 20\%$

1 = $\geq 10\%$ and $<20\%$

0 = $<10\%$

4. Rebound Pattern (daily):

2 = bullish rejection / bounce detected

0 = otherwise

5. Volume Confirmation:

1 = volume of latest candle $>$ 20-day average volume

0 = otherwise

6. Liquidity Bonus:

1 = AvgDollarVolume20 $\geq 20M$

0 = otherwise

7. Output File

Mandatory: CSV file

Optional: XLSX (only if included in same price)

Columns required:

- Symbol
- Close
- Support
- Resistance
- SupportDistance
- ResistanceDistance
- Amplitude
- Score
- CandidateFlag (TRUE/FALSE)

8. Logging

Log file required:

- Start time / End time
- Number of symbols scanned
- Number of candidates
- Errors per symbol

9. Error Handling

If data missing or invalid:

- Skip symbol
- Log issue
- Continue execution

10. Execution

Run once per day after US market close.

Manual or scheduled execution supported.

11. Out of Scope

No trade execution

No broker API integration for orders

No UI / dashboard

No database

No backtesting

No alerts (email/Telegram) unless specified separately